# Edge Detection

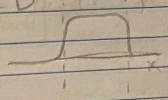
Edge: rapid Δ img intensity within small region

Causes:
- Depth Discontinuity (ex obj behind other obj)
- Surface Normal Discontinuity (light reflecting differently off of obj surface)
- Surface Reflectance Discont. (ex sticker has diff color)
- Illumination discont. (ex shadows)

Types of Edges
- Step edges
- Roof edge
- Line edges
- real edges

Edge Detector: invariant edge operator to produce:
- Edge position
- Edge magnitude
- orientation
- high detection rate
- good localization
- noise resilience

Using Gradients (1st derivative)
ex 1D edge edge



local extrema of 1st derivative indicate edges
→ take abs → height of peak := strength

w/ 2 dimensions
- partial derivatives: $\nabla I = \left(\frac{\partial I}{\partial x}, \frac{\partial I}{\partial y}\right)$
- magnitude $S = \|\nabla I\|$
- orientation $\theta = \tan^{-1}\left(\frac{\partial I}{\partial y} / \frac{\partial I}{\partial x}\right)$

Discretely...
- derivatives are finite differences

$\frac{\partial I}{\partial x} \approx \frac{1}{2\varepsilon}((I_{i+1,j+1} - I_{i,j+1}) + (I_{i+1,j} - I_{i,j}))$

$\frac{\partial I}{\partial y} \approx \frac{1}{2\varepsilon}((I_{i+1,j+1} - I_{i+1,j}) + (I_{i,j+1} - I_{i,j}))$

- can be implemented as convolution

$\frac{\partial}{\partial x} \approx \frac{1}{2\varepsilon}\begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix}$ $\frac{\partial}{\partial y} \approx \frac{1}{2\varepsilon}\begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$

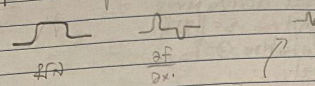
Note: convolution flips applied (!)

Comparing Gradient Operators

| Roberts (2x2) | Prewitt (3x3) | Sobel (3x3) | Sobel (5x5) |
|---|---|---|---|
| Good localization | | | Poor localization |
| Noise Sensitive | → | | Less noise Sensitive |
| Poor detection | | | Good detection |

Edge Thresholding
- Standard (Single Threshold T)
  - $\|\nabla I(x,y)\| < T$ → not edge
  - $\|\nabla I(x,y)\| \geq T$ → edge
- Hysteresis-Based (2 thresholds $T_0 < T_1$)
  - $\|\nabla I(x,y)\| < T_0$ → not edge
  - $\|\nabla I(x,y)\| \geq T_1$ → edge
  - $T_0 \leq \|\nabla I(x,y)\| < T_1$ → maybe edge if neighboring pixel is edge

Laplacian Operator (2nd derivative)



$\frac{\partial f}{\partial x}$ $\frac{\partial^2 f}{\partial x^2}$

zero-crossings indicate edges

Laplacian: sum of pure sec. der.

$\nabla^2 I = \frac{\partial^2 I}{\partial x^2} + \frac{\partial^2 I}{\partial y^2}$

- doesn't provide direction or orientation
- convolution mask:

$\nabla^2 = \frac{1}{\varepsilon^2}\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ or $\nabla^2 = \frac{1}{6\varepsilon^2}\begin{bmatrix} 1 & 4 & 1 \\ 4 & -20 & 4 \\ 1 & 4 & 1 \end{bmatrix}$

doesn't catch edges at angle well (better)

Effects of Noise
- obscures edge → apply gaussian first

gradient → $-\nabla(n_\sigma * f) = \nabla(n_\sigma) * f$ (where $n_\sigma$ is gaussian) because D and $n_\sigma$ are linear

laplacian → similarly...

$\nabla^2(n_\sigma * f) = \nabla^2(n_\sigma) * f$

| Gradient | vs | Laplacian |
|---|---|---|
| location, mag, orient. | | only location |
| detection w/ thresholding | | detection w/ zero-crossing |
| non-linear | | linear |

Canny Edge Detection
- smooth image w/ 2D gaussian: $n_\sigma * I$
- compute img gradient using sobel operator: $\nabla n_\sigma * I$
- find gradient magnitude at each pixel: $\|\nabla n_\sigma * I\|$
- find gradient orientation at each pixel:

$\hat{n} = \frac{\nabla n_\sigma * I}{\|\nabla n_\sigma * I\|}$

- compute 1D laplacian along gradient direction $\hat{n}$ at each pixel

$\frac{\partial^2 (n_\sigma * I)}{\partial \hat{n}^2}$

Depending on $\sigma$ for gaussian
- higher $\sigma$ → sparser edges
- lower $\sigma$ → finer edges
- allows edge detection at different scales

$D^2_{\vec{v}} f = D_{\vec{v}}(D_{\vec{v}}(f)) = D_{\vec{v}}(\nabla f \cdot \vec{v}) = \nabla(\nabla f \cdot \vec{v}) \cdot \vec{v}$

$= \nabla((f_x, f_y) \cdot \vec{v}) \cdot \vec{v} = \nabla(v_x f_x + v_y f_y) \cdot \vec{v}$